

Demonstration of AttraCar: Multisensory In-Car VR with Thermal, Airflow, and Motion Feedback through Built-In Vehicle Systems

Dohyeon Yeo
HCIS Lab, GIST
Gwangju, Republic of Korea
ing.dohyeonyeo@gm.gist.ac.kr

Gwangbin Kim
HCIS Lab, GIST
Gwangju, Republic of Korea
gwangbin@gm.gist.ac.kr

Minwoo Oh
HCIS Lab, GIST
Gwangju, Republic of Korea
alsdn4435@gm.gist.ac.kr

Jeongju Park
HCIS Lab, GIST
Gwangju, Republic of Korea
jeongjupark@gm.gist.ac.kr

Bocheon Gim
HCIS Lab, GIST
Gwangju, Republic of Korea
gimkenny1999@gm.gist.ac.kr

Seongjun Kang
HCIS Lab, GIST
Gwangju, Republic of Korea
ksjryan0728@gm.gist.ac.kr

Ahmed Elsharkawy*
HCIS Lab, GIST
Gwangju, Republic of Korea
elsharkawy@gm.gist.ac.kr

SeungJun Kim†
HCIS Lab, GIST
Gwangju, Republic of Korea
seungjun@gist.ac.kr

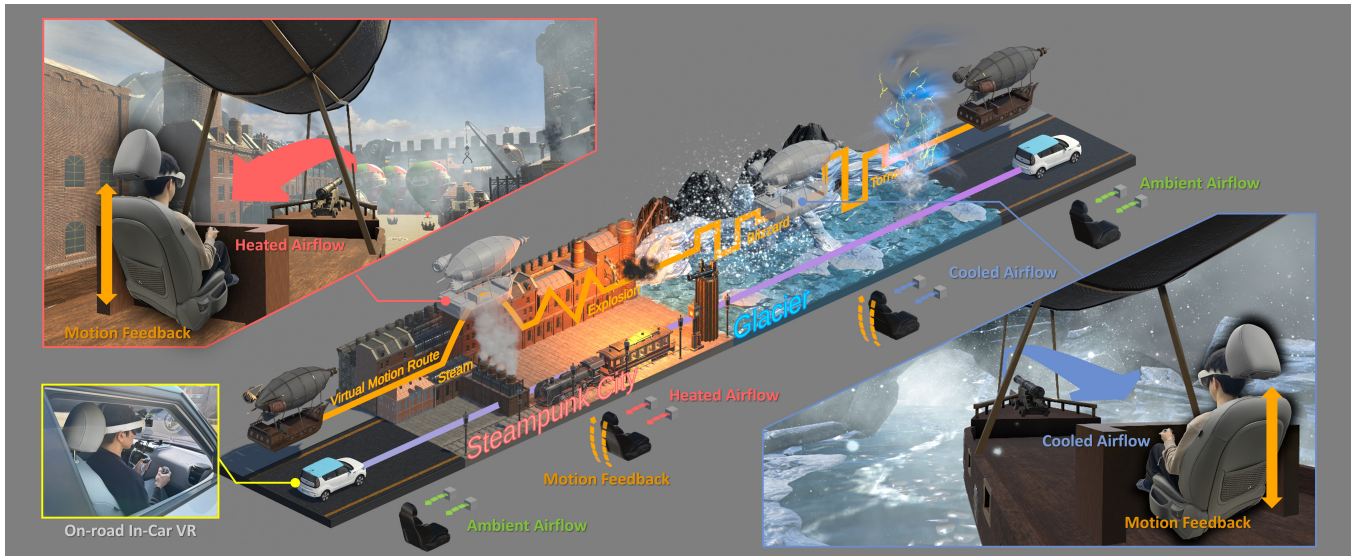


Figure 1: AttraCar immerses passengers in a VR airship journey through a warm *Steampunk City* and a cold *Glacier*, using the vehicle's HVAC and power seat to provide synchronized thermal, airflow, and motion feedback matching the virtual events.

Abstract

We introduce AttraCar, a multisensory in-car Virtual Reality (VR) platform that delivers thermal, airflow, and motion feedback using

*Department of Industrial Electronics and Control Engineering, Faculty of Electronic Engineering, Menoufa University, Egypt

†Corresponding Author.

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built-in vehicle systems. By controlling the Heating, Ventilation, and Air Conditioning (HVAC) system and the power seat, our platform enables the creation of multisensory, hardware-free in-car VR designed to enhance immersion and mitigate motion sickness. This demonstration allows attendees to experience these effects firsthand, showcasing a practical and scalable approach for future immersive automotive applications.

CCS Concepts

• Human-centered computing → Virtual reality.

Keywords

Virtual Reality, In-Car VR, Multisensory Feedback, Haptics

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1 INTRODUCTION

In-car Virtual Reality (VR) promises to transform passenger travel time into productive or entertaining experiences [7, 11, 12]. However, its adoption is hindered by challenges, most notably motion sickness caused by sensory mismatch and the impracticality of using external or wearable haptic devices in a confined vehicle cabin [3, 9, 10]. While many systems attempt to strictly synchronize virtual content with the car's physical motion to reduce sensory mismatch, this can limit the scope of immersive experiences [4–6].

To address these issues, we propose AttraCar, a novel platform that delivers rich multisensory feedback by repurposing the vehicle's own built-in systems. Instead of relying on external hardware, AttraCar leverages the Heating, Ventilation, and Air Conditioning (HVAC) system to provide dynamic airflow and thermal feedback, and the power seat to generate motion feedback (Figure 1). These modalities are synchronized with events in the virtual world, creating a perceptually coherent experience that enhances presence and reduces discomfort, even when the virtual narrative is decoupled from the vehicle's actual movement.

This demonstration showcases the effectiveness of our approach, allowing attendees to experience a multisensory in-car VR journey firsthand. Our work proposes that using existing vehicle components is a viable and scalable solution for creating immersive, comfortable, and hardware-free VR experiences for passengers.

2 SYSTEM DESIGN

The AttraCar platform is designed to seamlessly integrate VR content with the vehicle's native systems. The architecture, centered around a control unit, translates virtual events into physical feedback. A more detailed explanation about the technical implementation and user study result beyond this extended abstract can be found in Yeo et al [13].

2.1 Control System

A laptop running a Unity3D application serves as the main processor for the VR experience. It communicates with a custom control unit built on an Arduino Mega, which acts as the bridge to the vehicle's electronics (Figure 2). This unit interprets commands from the Unity scene and sends the appropriate signals to the car's sub-systems. For our demonstration, we use a KIA Soul EV, with the control unit installed discretely in the rear seat area.

2.2 Multisensory Feedback Implementation

We directly interface with the vehicle's control modules to generate haptic feedback. The intensity levels for each modality were calibrated through Just Noticeable Difference (JND) user experiments to ensure all feedback is perceptually meaningful [1, 8].

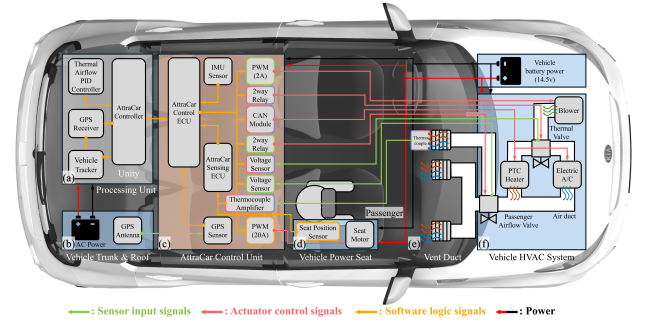


Figure 2: Overview of the AttraCar system architecture. (a) Unity-based in-car VR control module; (b) External power and GPS Antenna; (c) AttraCar control unit managing sensor and actuator signals; (d) Power seat module for vertical motion feedback; (e, f) Vehicle HVAC system for airflow and temperature control.

2.2.1 Airflow and Thermal Feedback. We bypass the default HVAC controls to directly manage the passenger-side vents. The airflow intensity is regulated by sending Pulse-Width Modulation (PWM) signals to the blower motor. Measurements showed maximum speeds of 8.0 m/s via the PTC Heater and 9.6 m/s via the Electric A/C. To prevent blower overload, the max airflow was limited to 7.6 m/s (95% PWM) for the PTC and A/C path.

For thermal feedback, we use relays and CAN bus signals to activate the PTC Heater or the electric A/C. A K-type thermocouple installed in the vent provides real-time temperature readings, allowing a PID controller to maintain the target temperature for heated (up to 50°C) or cooled (down to 15°C) airflow.

2.2.2 Motion Feedback. Vertical motion is generated using the passenger-side power seat. We control the seat's height adjustment motor with a high-current PWM signal. To ensure consistent motion speed regardless of passenger weight, we developed a regression model that calibrates the required PWM value based on the passenger's weight and the desired speed (up to 14 mm/s). This allows for smooth and predictable waveform-based motion cues, such as sine, triangle, or square waves, synchronized with virtual events [2].

$$Upward\ PWM = 0.69W + 12.68S + 0.95 \quad (1)$$

$$Downward\ PWM = -0.55W + 13.93S + 54.3 \quad (2)$$

where:

- W : Passenger weight (kg)
- S : Desired seat motion speed (mm/s)

3 DEMONSTRATION EXPERIENCE

Attendees will be invited to sit in the front passenger seat of our demonstration vehicle and wear a Meta Quest 3 headset. The demo places them in the role of a crew member on a pirate airship, embarking on a mission through dynamically changing environments.

The journey, illustrated in Figure 3, is designed to showcase the full range of multisensory feedback. The journey is designed to showcase the full range of multisensory feedback. The airship's



Figure 3: The AttraCar demonstration experience. (a) In the warm *Steampunk City* scenario, the system delivers heated airflow and synchronized motion to simulate the environment. (b) In the cold *Glacier* scenario, the feedback shifts to cooled airflow and different motion patterns to match the icy landscape, enhancing environmental presence.

horizontal travel is mapped to the real car’s movement, but its vertical motion and reactions to environmental events are rendered through the HVAC and power seat systems. The experience is divided into two main sections:

- **The Steampunk City:** This initial region is characterized by warm, industrial visuals. Attendees will feel heated airflow from the vents, simulating the heat from the airship’s steam engines. Events like ascending on a steam updraft are accompanied by gentle, sine-wave seat motion and increasing airflow. A sudden explosion event triggers a sharp, triangular motion wave and a burst of warm air.
- **The Glacier:** As the airship travels into a frigid, icy region, attendees will feel the ambient temperature drop as the HVAC system delivers cooled airflow. Here, events like navigating a blizzard are rendered with randomized, square-wave seat motions and strong gusts of cold wind, creating a starkly different and immersive sensory experience.

Throughout the 3-4 minute demonstration, participants will experience how these integrated feedback modalities work together to create a compelling and comfortable in-car VR experience, reducing the sensory conflicts that cause motion sickness.

4 CONCLUSION

This demonstration of AttraCar presents a novel and practical approach to in-car VR. By leveraging built-in vehicle systems to deliver rich, multisensory feedback (including thermal, airflow, and motion cues), our platform enhances user presence and comfort. By utilizing existing vehicle features without external or wearable devices, this approach makes multimodal VR accessible to passengers during travel and shows a practical path forward for in-vehicle entertainment.

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